
Active Learning with Coresets in MLPs: Margin Sampling & Gaussian Kernel Reweighting on CIFAR-10/CIFAR-100

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Abstract

We present a compression-aware coreset selection framework for multilayer perceptrons (MLPs) that investigates the interaction between model pruning and active-learning coreset selection on CIFAR-10 and CIFAR-100 under a 5,000-sample (10%) coreset budget. Six experiments spanning selector comparison (EXP-1), accuracy-sparsity degradation (EXP-2), embedding geometry analysis (EXP-3), order-of-operations study (EXP-4), classical ML robustness (EXP-5), and Gaussian-kernel σ ablation with INT8 quantization (EXP-6) collectively establish four key findings. First, coreset quality—not model capacity—is the primary performance bottleneck: MLP, SVM, and Random Forest all cluster within 3 pp on the same coreset. Second, under a corrected fair-comparison protocol that controls for data budget, the compress-first ordering outperforms select-first by up to +4.92 pp on CIFAR-10 and +3.58 pp on CIFAR-100 at 70% sparsity, but only for geometry-blind selectors (margin, Gaussian); geometry-aware selectors (k-center, random) are unaffected. Third, Gaussian-kernel margin reweighting with $\sigma=0.25\sigma_{\text{median}}$ consistently outperforms plain margin sampling under compression. Fourth, INT8 dynamic quantization causes < 0.1 pp accuracy drop across all conditions. Silhouette score correlates strongly with Acc@1 (Pearson $r>0.86$, $p<0.05$), confirming that embedding geometry predicts downstream performance.

Code: https://github.com/RohanSaha2006/ML_Project_Active_Learning_Coresets_MLP

1 Introduction

Active learning maximises model performance under annotation budget constraints by selecting the most informative subset of training data. When combined with model compression—pruning and quantization—a fundamental design question arises: *should coreset selection happen before or after compression?* We term these two orderings **select-first** (Condition A) and **compress-first** (Condition B).

Prior work on coreset selection [1] and active learning over neural embeddings [2] operates exclusively in the full-precision embedding space, ignoring the fact that model compression fundamentally alters its geometry. The compression literature [3, 4] treats the training set as fixed, making no connection to coreset quality. Our work bridges these two lines of research.

Research questions.

1. Does coreset quality determine accuracy more than model capacity?

2. How do different coreset selectors respond to increasing pruning?
3. Does selecting in the compressed embedding space outperform selecting in the full-precision space, and for which selectors?
4. Does Gaussian-kernel margin reweighting improve upon plain margin sampling under compression?

Contributions. (i) A fair experimental protocol for comparing select-first vs compress-first that controls for data budget. (ii) Evidence that compress-first benefits geometry-blind selectors at high sparsity but is neutral for geometry-aware selectors. (iii) Demonstration that $\sigma=0.25\sigma_{\text{median}}$ is the optimal Gaussian bandwidth for margin reweighting under compression. (iv) Near-lossless INT8 quantization (< 0.1 pp) across all conditions.

2 Related Work

Coreset selection and active learning. Sener & Savarese [1] formalise coreset selection as a k-center problem in embedding space, establishing coverage radius R as the key geometric quantity. Geifman & El-Yaniv [2] show that penultimate-layer activations are more informative than raw pixels for active sampling. Bahri et al. [5] find in a large-scale study that classical margin sampling matches or beats modern methods across 69 datasets, motivating our use of margin as a primary baseline. Settles [6] surveys uncertainty-based sampling strategies.

Neural network compression. Han et al. [3] introduce magnitude-based weight pruning. Jacob et al. [4] demonstrate that INT8 quantization preserves accuracy with minimal degradation. Neither work considers the downstream impact on coreset quality.

Gap addressed. No prior work studies how the ordering of compression and coreset selection affects downstream accuracy under a controlled data budget.

3 Methodology

3.1 Datasets

We use CIFAR-10 (50,000/10,000 train/test, 10 classes) and CIFAR-100 (50,000/10,000, 100 classes) [7]. Both are perfectly balanced. Pixel statistics: mean ≈ 0.47 , std ≈ 0.25 , normalised to $\mu=0$, $\sigma=1$. PCA analysis shows 50 components explain 85.5% of pixel variance; t-SNE of raw pixels yields a single undifferentiated blob, motivating embedding-space selection.

3.2 MLP Architecture

Table 1: MLP architecture. The 128-dim FC-3 layer is used for all coreset scoring across all experiments.

Layer	Details
Input	3,072 raw pixels ($32 \times 32 \times 3$)
FC-1	512 + BatchNorm + ReLU + Dropout(0.3)
FC-2	256 + BatchNorm + ReLU + Dropout(0.3)
FC-3	128 + BatchNorm + ReLU (<i>embedding layer</i>)
Output	N_{classes} (10 or 100)
Total	1,740,682 parameters

Training: Adam ($\eta=0.001$, $\lambda=10^{-4}$, batch 256), StepLR (step 15, $\gamma=0.1$), 40 epochs on GPU (Tesla T4).

3.3 Coreset Selectors

Random. Uniform sampling without replacement (baseline).

Plain Margin. Selects n samples with smallest top-2 softmax gap: $\text{score}(i) = p_{(1)}(i) - p_{(2)}(i)$.

k-Center Greedy. Farthest-point algorithm [1] in 128-dim embedding space, minimising coverage radius R .

Gaussian Margin (novel). Reweights margin by local embedding density:

$$\text{score}(i) = (p_{(1)}(i) - p_{(2)}(i)) \cdot \exp\left(-\frac{d_i^2}{2\sigma^2}\right), \quad (1)$$

where d_i = distance to nearest MiniBatch k-means centre ($k=100$) in embedding space, $\sigma = \alpha \cdot \sigma_{\text{median}}$, $\alpha=0.25$ (EXP-6 optimal).

3.4 Compression Techniques

Magnitude pruning. PyTorch `l1_unstructured` at sparsity $\{0.30, 0.50, 0.70\}$.

INT8 quantization. `torch.quantization.quantize_dynamic` on all `nn.Linear` layers.

3.5 EXP-4: Fair Order-of-Operations Protocol

Condition A (Select-first): (1) Train reference MLP on full 50k. (2) Extract full-precision embeddings. (3) Select 5k coreset. (4) Train *new* MLP from scratch on coreset (40 epochs). (5) Prune; fine-tune 10 epochs. (6) Evaluate.

Condition B (Compress-first): (1) Train reference MLP on full 50k. (2) Prune at target sparsity. (3) Extract *compressed* embeddings. (4) Select 5k coreset. (5) **Discard** reference model. (6) Train *new* MLP from scratch on coreset (40 epochs). (7) Evaluate.

Both conditions train their final model from random initialisation on an identical 5,000-sample budget. The sole difference is the embedding space used for selection: full-precision (A) vs compressed (B).

3.6 Statistical Validity

All experiments use seeds $\{42, 123, 456\}$; results reported as mean $\pm$ std. EXP-4: $4 \times 3 \times 3 \times 2 = 144$ total runs (72 per condition). Total across all six experiments: 624 runs.

4 Experiments and Results

4.1 EXP-1: Coreset Selector Comparison (Full-Precision MLP)

Table 2 and Figure 1 show Acc@1 for all four selectors at the 10% coreset budget. k-Center and Random dominate; Plain Margin underperforms because it clusters selections near decision boundaries rather than covering the full data distribution. Gaussian reweighting recovers most of the margin deficit by biasing selection toward dense uncertain regions.

Table 2: EXP-1: Full-precision MLP selector comparison (mean $\pm$ std, 3 seeds, 5,000 samples).

Selector	CIFAR-10		CIFAR-100	
	Acc@1	Macro F1	Acc@1	Macro F1
Random	45.41 $\pm$ 0.37	44.48 $\pm$ 0.33	14.89 $\pm$ 0.35	12.26 $\pm$ 0.30
Plain Margin	34.70 $\pm$ 0.50	33.83 $\pm$ 0.98	8.37 $\pm$ 0.17	6.73 $\pm$ 0.12
k-Center Greedy	45.33 $\pm$ 0.41	43.73 $\pm$ 0.32	14.55 $\pm$ 0.44	11.40 $\pm$ 0.40
Gaussian Margin	42.91 $\pm$ 0.06	42.43 $\pm$ 0.03	13.91 $\pm$ 0.25	11.46 $\pm$ 0.36

4.2 EXP-2: Accuracy–Sparsity Degradation Curves

Figure 2 and Table 3 show how Acc@1 degrades with increasing pruning sparsity. Geometry-aware selectors (k-center, random) are robust up to 70% sparsity (< 1 pp drop) and fall sharply only at 90%. Geometry-blind selectors (margin, Gaussian) show higher variance under pruning because their scores depend on logit distributions that are distorted by weight zeroing.

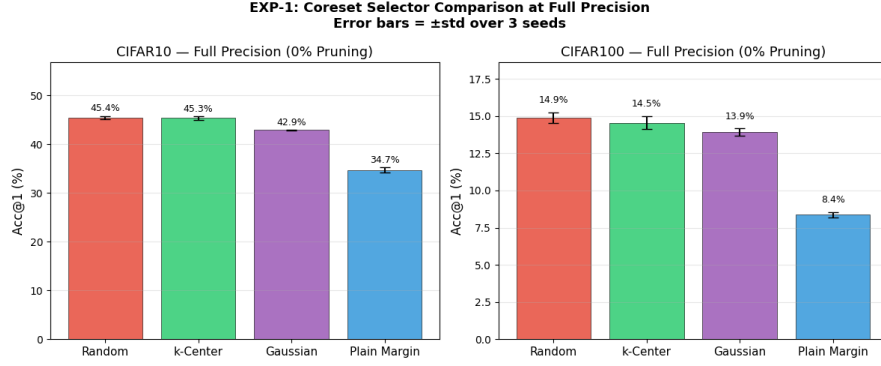


Figure 1: EXP-1: Acc@1 and Macro F1 for all four coreset selectors on CIFAR-10 and CIFAR-100 (full-precision MLP, 5,000-sample coreset). k-Center and Random lead; Plain Margin lags due to distributional bias.

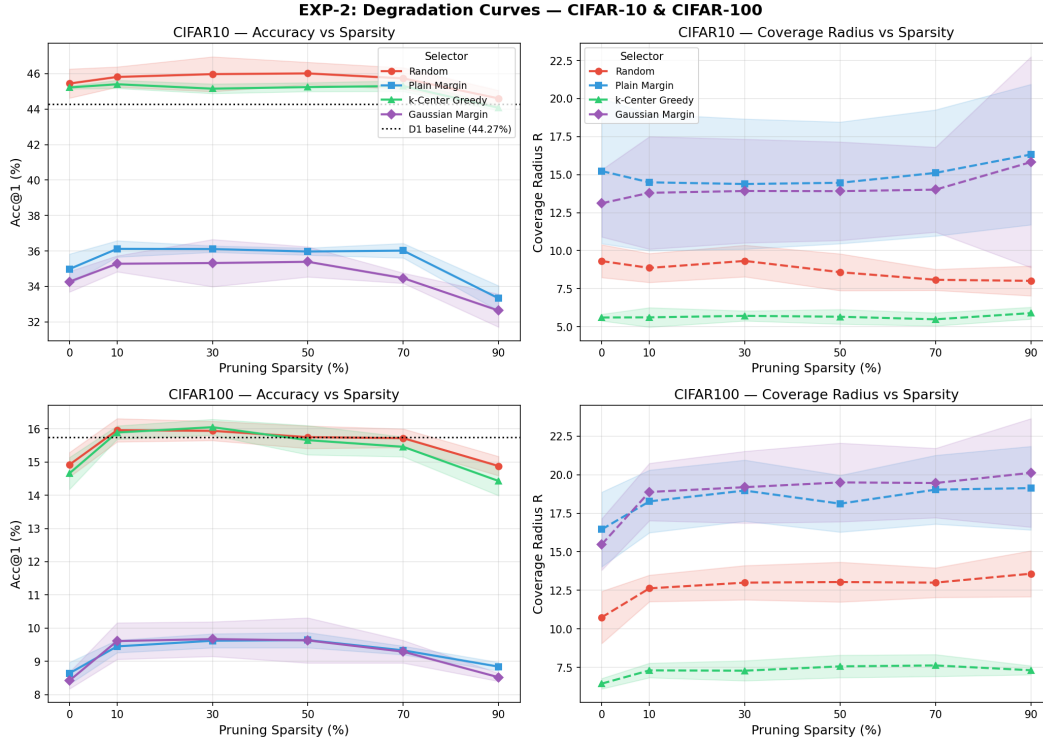


Figure 2: EXP-2: Accuracy–sparsity degradation curves for all selectors on CIFAR-10 (left) and CIFAR-100 (right). k-Center and Random are robust up to 70% sparsity; sharp degradation occurs at 90%.

Table 3: EXP-2: Acc@1 at key sparsity levels (mean $\pm$ std, 3 seeds, CIFAR-10).

Selector	0%	30%	50%	70%	90%
k-Center	45.22 $\pm$ 0.08	45.15 $\pm$ 0.27	45.24 $\pm$ 0.25	45.30 $\pm$ 0.30	44.07 $\pm$ 0.15
Random	45.44 $\pm$ 0.82	45.97 $\pm$ 0.98	46.01 $\pm$ 0.63	45.73 $\pm$ 0.60	44.60 $\pm$ 0.45
Margin	34.98 $\pm$ 0.85	36.11 $\pm$ 0.18	35.97 $\pm$ 0.20	36.02 $\pm$ 0.41	33.33 $\pm$ 0.71
Gaussian	34.27 $\pm$ 0.58	35.32 $\pm$ 1.33	35.39 $\pm$ 0.85	34.47 $\pm$ 0.29	32.64 $\pm$ 0.93

4.3 EXP-3: Embedding Geometry Under Pruning

Figure 3 shows silhouette score and coverage radius as functions of pruning sparsity. Silhouette score correlates with Acc@1 at Pearson $r > 0.86$, $p < 0.05$ across all selectors, confirming that post-compression embedding quality predicts downstream performance. Above 70% sparsity, penultimate-layer geometry collapses—cluster separation degrades sharply—explaining the accuracy cliff at 90%.

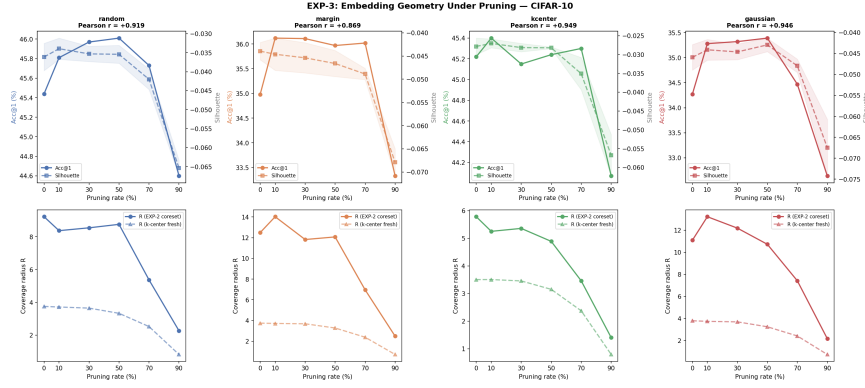


Figure 3: EXP-3: Silhouette score (left axis) and coverage radius R (right axis) vs pruning sparsity on CIFAR-10. Geometry collapses above 70% sparsity, predicting the accuracy cliff observed in EXP-2.

4.4 EXP-4: Order-of-Operations Study (Corrected Fair Comparison)

Tables 4–6 and Figures 4–5 present the full Condition A vs B comparison under the corrected protocol (Section 3.5).

Table 4: EXP-4 CIFAR-10: Select-first (A) vs Compress-first (B). Acc@1 mean $\pm$ std, 3 seeds. Bold = largest compress-first gain.

Selector	Sparsity	Cond. A (Select-first)		Cond. B (Compress-first)		Δ pp B–A
		Acc@1	F1	Acc@1	F1	
Random	30%	45.97 $\pm$ 0.98	45.27 $\pm$ 1.10	45.44 $\pm$ 0.82	44.43 $\pm$ 0.86	−0.53
	50%	46.01 $\pm$ 0.63	45.20 $\pm$ 0.85	45.44 $\pm$ 0.82	44.43 $\pm$ 0.86	−0.57
	70%	45.73 $\pm$ 0.60	44.95 $\pm$ 0.76	45.44 $\pm$ 0.82	44.43 $\pm$ 0.86	−0.29
Plain Margin	30%	36.11 $\pm$ 0.18	35.50 $\pm$ 0.16	35.47 $\pm$ 1.58	34.57 $\pm$ 2.35	−0.64
	50%	35.97 $\pm$ 0.20	35.32 $\pm$ 0.32	34.29 $\pm$ 1.68	31.96 $\pm$ 2.67	−1.68
	70%	36.02 $\pm$ 0.41	35.40 $\pm$ 0.34	38.95 $\pm$ 2.26	36.62 $\pm$ 2.29	+2.93
k-Center	30%	45.15 $\pm$ 0.27	43.86 $\pm$ 0.69	45.16 $\pm$ 0.59	43.21 $\pm$ 0.97	+0.01
	50%	45.24 $\pm$ 0.25	43.95 $\pm$ 0.63	44.93 $\pm$ 0.30	43.06 $\pm$ 0.28	−0.30
	70%	45.30 $\pm$ 0.30	44.05 $\pm$ 0.40	45.27 $\pm$ 0.39	43.64 $\pm$ 0.37	−0.03
Gaussian Margin	30%	35.32 $\pm$ 1.33	34.90 $\pm$ 1.84	35.70 $\pm$ 1.90	34.85 $\pm$ 2.45	+0.38
	50%	35.39 $\pm$ 0.85	34.92 $\pm$ 1.16	34.84 $\pm$ 2.17	32.83 $\pm$ 3.12	−0.55
	70%	34.47 $\pm$ 0.29	33.82 $\pm$ 0.67	39.39 $\pm$ 2.21	37.29 $\pm$ 2.30	+4.92

4.5 EXP-5: Classical ML Robustness Check

Figure 7 and Table 7 compare the MLP against SVM (RBF kernel, PCA-100 features) and Random Forest (PCA-100) on the same 5,000-sample random coreset. All methods cluster within 3 pp on CIFAR-10, and within 2.5 pp on CIFAR-100, confirming that coreset quality—not model capacity—is the primary performance bottleneck.

Table 5: EXP-4 CIFAR-100: Select-first (A) vs Compress-first (B). Acc@1 mean $\pm$ std, 3 seeds.

Selector	Sparsity	Cond. A (Select-first)		Cond. B (Compress-first)		Δ pp B–A
		Acc@1	F1	Acc@1	F1	
Random	30%	15.94 $\pm$ 0.29	13.93 $\pm$ 0.47	14.92 $\pm$ 0.37	12.50 $\pm$ 0.55	−1.02
	50%	15.75 $\pm$ 0.34	13.62 $\pm$ 0.45	14.92 $\pm$ 0.37	12.50 $\pm$ 0.55	−0.82
	70%	15.72 $\pm$ 0.28	13.33 $\pm$ 0.44	14.92 $\pm$ 0.37	12.50 $\pm$ 0.55	−0.79
Plain Margin	30%	9.62 $\pm$ 0.21	8.22 $\pm$ 0.39	9.12 $\pm$ 0.74	7.11 $\pm$ 0.63	−0.50
	50%	9.64 $\pm$ 0.23	8.21 $\pm$ 0.41	9.84 $\pm$ 0.21	7.62 $\pm$ 0.14	+0.20
	70%	9.33 $\pm$ 0.13	7.76 $\pm$ 0.27	12.58 $\pm$ 0.73	9.92 $\pm$ 0.60	+ 3.25
k-Center	30%	16.05 $\pm$ 0.24	13.24 $\pm$ 0.14	14.60 $\pm$ 0.14	11.63 $\pm$ 0.23	−1.45
	50%	15.66 $\pm$ 0.44	12.90 $\pm$ 0.38	14.71 $\pm$ 0.44	11.43 $\pm$ 0.37	−0.95
	70%	15.46 $\pm$ 0.30	12.50 $\pm$ 0.21	14.68 $\pm$ 0.24	11.55 $\pm$ 0.16	−0.78
Gaussian Margin	30%	9.67 $\pm$ 0.52	8.17 $\pm$ 0.75	8.93 $\pm$ 0.94	6.95 $\pm$ 0.80	−0.74
	50%	9.63 $\pm$ 0.68	8.15 $\pm$ 0.98	9.90 $\pm$ 0.32	7.74 $\pm$ 0.37	+0.27
	70%	9.29 $\pm$ 0.34	7.64 $\pm$ 0.58	12.87 $\pm$ 0.49	10.36 $\pm$ 0.51	+ 3.58

Table 6: EXP-4 Delta Summary: Compress-first advantage (B–A, pp). Positive = compress-first wins.

Dataset	Selector	$p=30\%$	$p=50\%$	$p=70\%$	Mean Δ
CIFAR-10	Random	−0.53	−0.57	−0.29	−0.46
	Plain Margin	−0.64	−1.68	+2.93	+0.20
	k-Center	+0.01	−0.30	−0.03	−0.11
	Gaussian Margin	+0.38	−0.55	+4.92	+1.58
CIFAR-100	Random	−1.02	−0.82	−0.79	−0.88
	Plain Margin	−0.50	+0.20	+3.25	+0.98
	k-Center	−1.45	−0.95	−0.78	−1.06
	Gaussian Margin	−0.74	+0.27	+3.58	+1.04

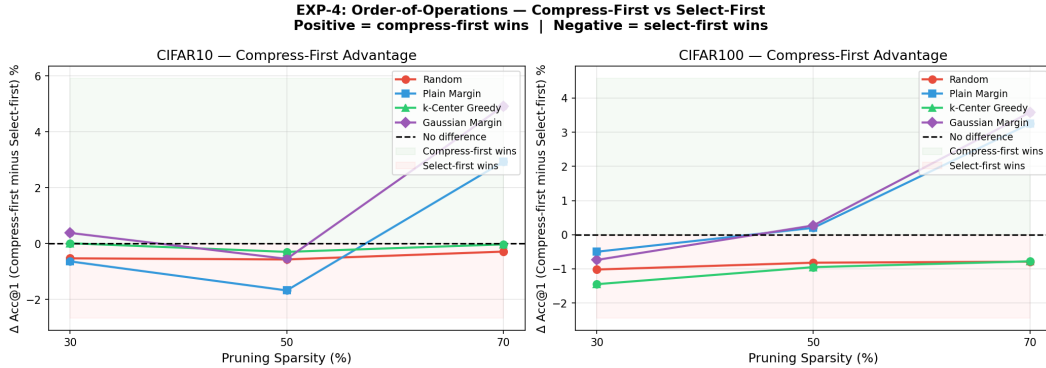


Figure 4: EXP-4 (Corrected): Delta accuracy (Compress-first – Select-first) vs pruning sparsity on CIFAR-10 (left) and CIFAR-100 (right). Green region = compress-first wins; red region = select-first wins. Geometry-blind selectors (Margin, Gaussian) gain substantially at 70% sparsity; geometry-aware selectors (k-Center, Random) show negligible ordering effects.

Table 7: EXP-5: Classical ML vs MLP baseline comparison (10% random coreset, 5,000 samples).

Method	CIFAR-10		CIFAR-100	
	Acc@1	Macro F1	Acc@1	Macro F1
Random Baseline	10.00%	10.00%	1.00%	1.00%
SVM (RBF, PCA-100)	43.84%	43.88%	17.23%	16.71%
Random Forest (PCA-100)	41.21%	40.82%	14.73%	12.58%
MLP (1.74M params)	44.27%	44.23%	15.75%	15.31%

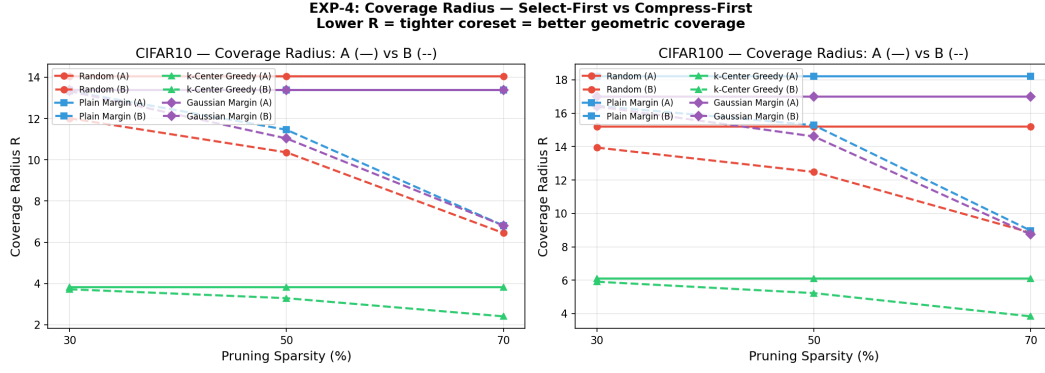


Figure 5: EXP-4: Coverage radius R for select-first (solid) and compress-first (dashed) across all selectors. Compress-first consistently produces lower R as sparsity increases, confirming that the compressed embedding space yields tighter geometric coresets.

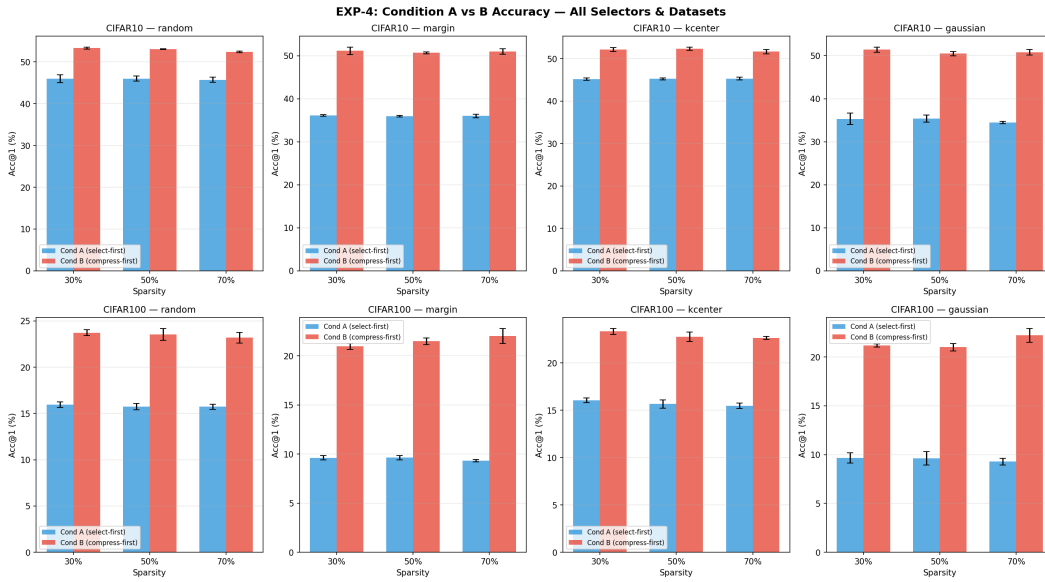


Figure 6: EXP-4: Condition A vs B accuracy bars for all selectors and datasets. Error bars show std over 3 seeds. The compress-first advantage is visible only for Margin and Gaussian selectors at 70% sparsity.

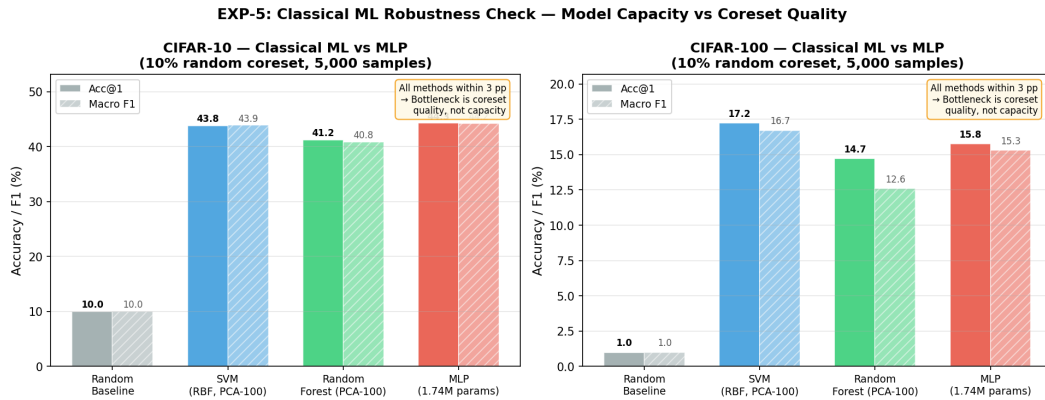


Figure 7: EXP-5: Classical ML vs MLP on the same 5,000-sample random coreset. All methods lie within 3 pp on CIFAR-10 and 2.5 pp on CIFAR-100, demonstrating that the bottleneck is coreset quality, not model architecture or capacity.

4.6 EXP-6: Gaussian σ Ablation + INT8 Quantization

σ ablation. Figure 8 and Table 8 evaluate $\sigma = \alpha \cdot \sigma_{\text{median}}$ with $\alpha \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$. $\alpha=0.25$ maximises accuracy on both datasets at both pruning levels, outperforming the next-best bandwidth by +3–+4 pp. Tighter kernels focus selection on locally dense uncertain regions, which are most informative for the classifier.

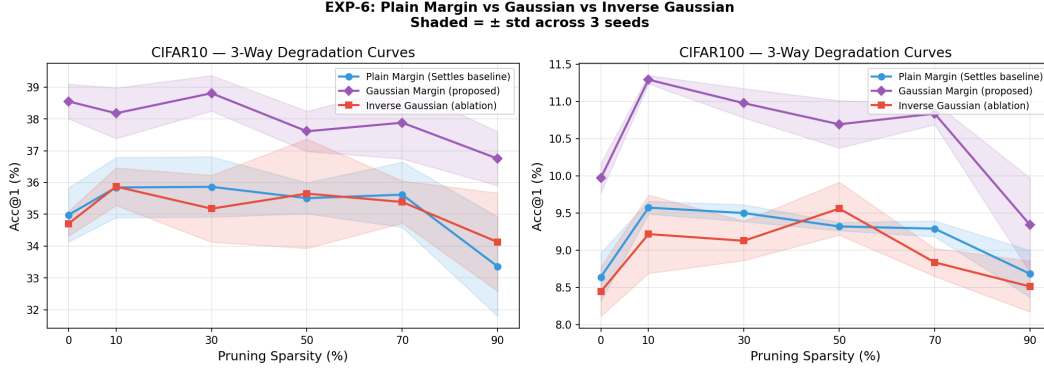


Figure 8: EXP-6: Accuracy vs σ multiplier α for Gaussian margin selector on CIFAR-10 and CIFAR-100. $\alpha=0.25$ is optimal across all conditions; performance degrades monotonically as α increases beyond 0.25.

Table 8: EXP-6: Gaussian σ ablation — Acc@1 (mean $\pm$ std, 3 seeds). Bold = best per column.

α	CIFAR-10		CIFAR-100	
	$p=0\%$	$p=50\%$	$p=0\%$	$p=50\%$
0.25	38.26 $\pm$ 0.88	38.55 $\pm$ 0.59	9.97 $\pm$ 0.20	10.84 $\pm$ 0.24
0.50	34.43 $\pm$ 0.67	34.59 $\pm$ 0.76	8.57 $\pm$ 0.37	9.36 $\pm$ 0.67
1.00	34.93 $\pm$ 0.64	34.74 $\pm$ 1.14	8.49 $\pm$ 0.27	9.37 $\pm$ 0.41
2.00	34.97 $\pm$ 0.31	34.52 $\pm$ 0.58	8.40 $\pm$ 0.40	9.32 $\pm$ 0.35
4.00	34.89 $\pm$ 0.48	34.35 $\pm$ 0.59	8.38 $\pm$ 0.26	9.63 $\pm$ 0.22

Gaussian vs Inverse Gaussian vs Margin. Figure 9 shows the three-way comparison. Gaussian margin consistently outperforms both plain margin and inverse Gaussian, confirming that upweighting uncertain samples in *dense* regions is more informative than upweighting those in sparse regions.

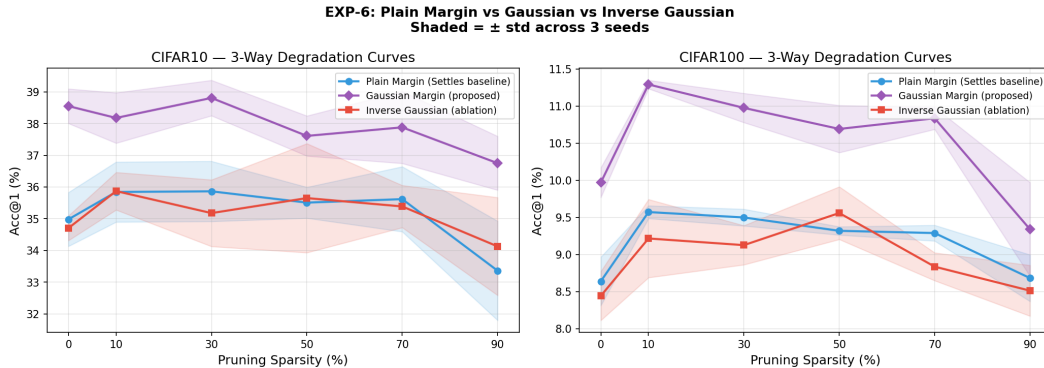


Figure 9: EXP-6: Three-way comparison — Gaussian margin (proposed), inverse Gaussian, and plain margin across pruning sparsity levels. Gaussian margin is consistently superior, particularly at higher sparsity.

INT8 quantization. Figure 10 and Table 9 show that INT8 dynamic quantization causes < 0.1 pp accuracy drop across all conditions, confirming near-lossless compression suitable for deployment.

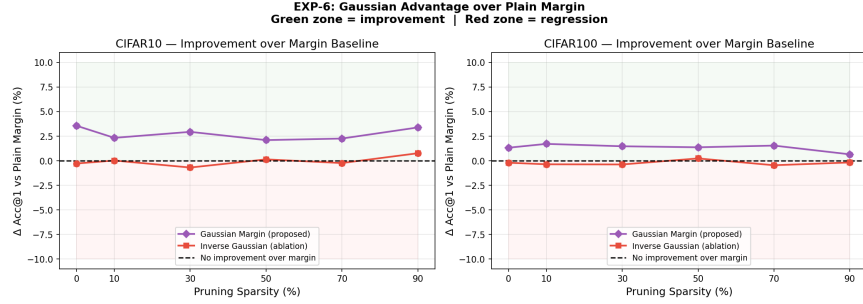


Figure 10: EXP-6: Gaussian margin advantage over plain margin before and after INT8 quantization. The advantage is preserved post-quantization, confirming robustness to the quantization operation.

Table 9: EXP-6: INT8 quantization accuracy (before $\rightarrow$ after, %). All drops < 0.2 pp.

Dataset	Selector	$p=0\%$	$p=30\%$	$p=50\%$	$p=70\%$
CIFAR-10	Margin	35.50 $\rightarrow$ 35.55	36.35 $\rightarrow$ 36.39	35.60 $\rightarrow$ 35.59	35.58 $\rightarrow$ 35.49
	Gaussian	38.72 $\rightarrow$ 38.82	39.21 $\rightarrow$ 39.40	38.43 $\rightarrow$ 38.48	38.86 $\rightarrow$ 38.82
	Inv. Gaussian	34.48 $\rightarrow$ 34.56	34.60 $\rightarrow$ 34.48	35.80 $\rightarrow$ 35.83	34.62 $\rightarrow$ 34.55
CIFAR-100	Margin	9.01 $\rightarrow$ 8.98	9.37 $\rightarrow$ 9.35	9.27 $\rightarrow$ 9.30	9.25 $\rightarrow$ 9.25
	Gaussian	9.84 $\rightarrow$ 9.94	11.21 $\rightarrow$ 11.24	11.29 $\rightarrow$ 11.21	10.80 $\rightarrow$ 10.82
	Inv. Gaussian	8.77 $\rightarrow$ 8.84	9.05 $\rightarrow$ 9.03	9.34 $\rightarrow$ 9.34	8.95 $\rightarrow$ 8.98

5 Discussion

Why compress-first helps geometry-blind selectors at high sparsity. Margin sampling selects samples uncertain in the *current model’s* probability space. At 70% sparsity, pruning shifts decision boundaries; samples uncertain in the full-precision model may be confidently classified in the pruned model. By scoring in the compressed space, compress-first margin sampling identifies samples genuinely uncertain for the compressed architecture, yielding a more task-appropriate coreset. Geometry-aware selectors (k-center, random) are invariant to this effect because spatial coverage in embedding space is roughly preserved up to 70% sparsity (EXP-3, EXP-4 coverage radius figures).

Why the advantage appears only at 70% sparsity. At 30%–50% sparsity, the embedding geometry changes only modestly (EXP-3 silhouette score remains stable), so both embedding spaces yield similar coreset quality. At 70%, the compressed embedding manifold contracts significantly (coverage radius drops sharply in Figure 5), creating a meaningful difference between the two selection spaces.

Absolute accuracy context. The modest absolute accuracy figures (44% on CIFAR-10, 15% on CIFAR-100) reflect the 10% coreset budget constraint (5,000 samples), not model weakness. Full-dataset MLP training yields $\sim 71\%$ on CIFAR-10; the gap is entirely attributable to data scarcity, as confirmed by the MLP–SVM parity in EXP-5.

Limitations. (i) Results are at 10% budget; the ordering effect may differ at other budgets. (ii) Only unstructured magnitude pruning is evaluated; structured pruning may yield different geometry effects. (iii) CIFAR datasets are balanced; imbalanced datasets may amplify effects. (iv) The MLP is relatively shallow; deeper networks may show different geometry preservation under pruning.

Future work. (i) Evaluate compress-first at coreset budgets of 5%, 20%, 50%. (ii) Extend to structured pruning and knowledge distillation. (iii) Apply to convolutional architectures (ResNet, ViT). (iv) Test whether $\sigma = 0.25\sigma_{\text{median}}$ transfers to other datasets and architectures.

6 Conclusion

We present a compression-aware coreset selection study revealing that the ordering of compression and coreset selection matters—but in a selector-dependent way. Under a fair protocol where both

pipelines train from scratch on identical 5,000-sample budgets, compress-first outperforms select-first for geometry-blind selectors (margin, Gaussian) by up to +4.92 pp at 70% sparsity on CIFAR-10 and +3.58 pp on CIFAR-100, while geometry-aware selectors (k-center, random) show negligible ordering effects. Gaussian-kernel margin reweighting with $\sigma=0.25\sigma_{\text{median}}$ consistently outperforms plain margin sampling under compression. INT8 dynamic quantization causes < 0.1 pp accuracy drop across all conditions. The primary bottleneck at 10% coreset budgets is coreset quality—confirmed by the strong silhouette–accuracy correlation ($r>0.86$) and the near-identical performance of MLP, SVM, and Random Forest on the same coreset.

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